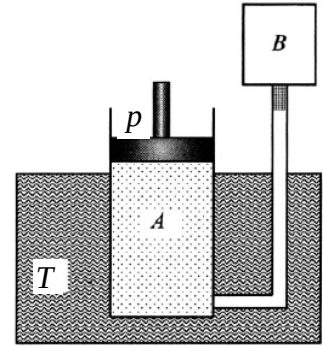


University Physics Quiz 6-3

Name: _____ SID: _____ School/Department: _____

As shown in the accompanying figure, container A of variable volume V controlled by a frictionless piston, is immersed in a heat bath at temperature T . The container is connected by a pipe with a porous plug to another insulated container B of fixed volume. Container A is initially occupied with ideal gas at pressure p while container B is initially evacuated. The gas is allowed to flow through the plug into the container B and the pressure in the container A is maintained at p by pushing the piston. Show that when the pressure of the gas in container B reaches p , the temperature of it is



$$T_B = \frac{c_p}{c_v} T$$

where c_p and c_v are isobaric and isochoric molar heat capacities of the ideal gas.

Solution: When the piston is pushed, the work done to the system is $\Delta W = p\Delta V$, where ΔV is the reduced volume of the gas in container A. The part of gas of Δn mole in container A (of volume ΔV) will go through the porous plug into container B occupying volume $\Delta V'$ with pressure p' , the work by the gas is $\Delta W' = (p - p')\Delta V'$. Therefore, using the first law, we have

$$p\Delta V = \Delta n c_v (T_B - T) + (p - p')\Delta V'$$

From the equation of state, we have $p\Delta V = \Delta n RT$, $p'\Delta V' = \Delta n RT_B$. Substituting them into above equation, we obtain

$$\Delta n RT = \Delta n c_v (T_B - T) + \left(\frac{p}{p'} - 1 \right) \Delta n RT_B \quad \text{or} \quad (c_v + R)T = \left[c_v + \left(\frac{p}{p'} - 1 \right) R \right] T_B.$$

Thus

$$T_B = \frac{(c_v + R)T}{\left[c_v + \left(\frac{p}{p'} - 1 \right) R \right]}.$$

When $p' = p$, we obtain $T_B = \frac{c_p}{c_v} T$.

2. In an ideal gas of molecular mass m , it is found that the speeds v_1 and v_2 have the equal value of the Maxwell speed distribution function. Find the temperature of the gas.

Solution:

$$f(v) = 4\pi n_0 v^2 \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-\frac{mv^2}{2k_B T}}.$$

$$f(v_1) = f(v_2) \Rightarrow v_1^2 e^{-\frac{mv_1^2}{2k_B T}} = v_2^2 e^{-\frac{mv_2^2}{2k_B T}}.$$

$$\frac{v_1^2}{v_2^2} = e^{\frac{m(v_1^2 - v_2^2)}{2k_B T}},$$

$$2 \ln \frac{v_1}{v_2} = \frac{m(v_1^2 - v_2^2)}{2k_B T}, \quad T = \frac{m(v_1^2 - v_2^2)}{4k_B \ln \frac{v_1}{v_2}}.$$

3. A definite amount of gas at temperature T_0 initially is confined to volume V_0 in a partitioned insulated rigid cylinder. The unoccupied part, also with volume V_0 is initially evacuated. Suppose the gas expands spontaneously and freely into the unoccupied part, and eventually comes to the equilibrium.

(a) What is the change in internal energy in the process?

(b) Show that the final temperature of the gas is

$$T_f = T_0 + \int_{V_0}^{2V_0} \frac{1}{C_V} \left[p - T \left(\frac{\partial p}{\partial T} \right)_V \right] dV.$$

(c) Show that $T_f = T_0$ for an ideal gas and $T_f < T_0$ for a gas that obeys the van der Waals equation. Explain the results in terms of the kinetic theory of molecules.

Solution: (a) In the process, no work is done and no heat is transferred. Thus the internal energy is unchanged, $\Delta U = 0$.

To deal with the problem, we consider the quantity $(\partial T / \partial V)_U$. Using the cyclic chain rule, we have

$$\left(\frac{\partial T}{\partial V} \right)_U = - \frac{\left(\frac{\partial U}{\partial V} \right)_T}{\left(\frac{\partial U}{\partial T} \right)_V} = - \frac{\left(\frac{\partial U}{\partial V} \right)_T}{C_V}.$$

From

$$dU = TdS - pdV = T \left(\frac{\partial S}{\partial T} \right)_V + \left[T \left(\frac{\partial S}{\partial V} \right)_T - p \right] dV,$$

we have

$$\left(\frac{\partial U}{\partial V} \right)_T = T \left(\frac{\partial S}{\partial V} \right)_T - p.$$

From $dF = -SdT - pdV$, we have the Maxwell relation $\left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial p}{\partial T} \right)_V$. Then, we have

$$\left(\frac{\partial U}{\partial V} \right)_T = T \left(\frac{\partial p}{\partial T} \right)_V - p.$$

Thus, we have

$$dT = \left(\frac{\partial T}{\partial V} \right)_U dV + \left(\frac{\partial T}{\partial U} \right)_V dU = \left(\frac{\partial T}{\partial V} \right)_U dV = \frac{1}{C_V} \left(p - T \left(\frac{\partial p}{\partial T} \right)_V \right) dV,$$

and

$$\int_{T_0}^{T_f} dT = \int_{V_0}^{2V_0} \frac{1}{C_V} \left[p - T \left(\frac{\partial p}{\partial T} \right)_V \right] dV, \quad T_f = T_0 + \int_{V_0}^{2V_0} \frac{1}{C_V} \left[p - T \left(\frac{\partial p}{\partial T} \right)_V \right] dV$$

(c) For an ideal gas

$$p - T \left(\frac{\partial p}{\partial T} \right)_V = \frac{nRT}{V} - T \frac{nR}{V} = 0,$$

Then $T_f = T_0$.

For the van der Waals gas, $p = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2}$,

$$p - T \left(\frac{\partial p}{\partial T} \right)_V = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2} - T \frac{nR}{V - nb} = - \frac{n^2 a}{V^2} < 0,$$

Then $T_f < T_0$.

For the ideal gas, as the gas expands, the kinetic energy of the molecules is unchanged and so is the temperature.

For the van der Waals gas, as the gas expands, the spacing among molecules increases. Thus the attractive (negative) interaction potential energy increases and the translational kinetic energy of the molecules decreases and therefore the temperature drops.

4. (a) There is an amount of matter in gas-liquid equilibrium. Assume that the gas phase of the matter behaves as an ideal gas and the volume of the matter in liquid phase is negligible. Show that the expansion coefficient of the matter in the gas-liquid equilibrium is given by

$$\frac{1}{V} \frac{dV}{dT} = \frac{1}{T} \left(1 - \frac{L}{RT} \right)$$

where L is the molar latent heat.

(b) Show that the change in molar internal energy in a phase transition can be expressed as

$$\Delta u = L \left(1 - \frac{p}{T} \frac{dT}{dp} \right).$$

Solution: (a) The volume of the matter is approximately the volume of the part of matter in gas phase. Then, the molar volume of the matter can be written as $v = v(T, p)$. Thus

$$\frac{1}{v} \frac{dv}{dT} = \frac{1}{v} \left(\left(\frac{\partial v}{\partial T} \right)_p + \left(\frac{\partial v}{\partial p} \right)_T \frac{dp}{dT} \right). \quad (1)$$

Since the gas phase of the matter behaves as an ideal gas, then $pv = xRT$, where x is the molar fraction of the matter in gas phase. Therefore

$$\frac{1}{v} \left(\frac{\partial v}{\partial T} \right)_p = \frac{1}{T}, \quad \frac{1}{v} \left(\frac{\partial v}{\partial p} \right)_T = -\frac{1}{p}. \quad (2)$$

Using the Clapeyron equation, $\frac{dp}{dT} = \frac{L}{T(V_m^g - V_m^l)} \approx \frac{L}{TV_m^g}$. Noting that $V_m^g = \frac{RT}{p}$, we have

$$\frac{dp}{dT} = \frac{Lp}{RT^2}. \quad (3)$$

Substituting (2) and (3) into (1), we finally obtain

$$\frac{1}{v} \frac{dv}{dT} = \frac{1}{T} \left(1 - \frac{L}{RT} \right).$$

(b) The condition for the phase equilibrium is

$$\mu_\alpha(T, p) = \mu_\beta(T, p)$$

where μ represents the chemical potentials of the two phases.

Since $\mu = u + pv - Ts$ where v is the molar volume and s is the molar entropy, we have

$$u_\alpha + pv_\alpha - Ts_\alpha = u_\beta + pv_\beta - Ts_\beta.$$

Thus $\Delta u = u_\beta - u_\alpha = T(s_\beta - s_\alpha) - p(v_\beta - v_\alpha) = L - p(v_\beta - v_\alpha)$.

Using the Clausius-Clapeyron equation $\frac{dp}{dT} = \frac{L}{T(v_\beta - v_\alpha)}$, we have $v_\beta - v_\alpha = \frac{L}{T} \frac{dT}{dp}$.

Finally, we obtain $\Delta u = L \left(1 - \frac{p}{T} \frac{dT}{dp} \right)$.